

Graph Caption Framing Teacher Notes

TOK Cognitive Science Activity Bank • Knowledge and Language • 20-30 min

Category	Knowledge and Language	Time	20-30 min
Difficulty	Easy	ID	graph-caption-framing

Big idea: Data needs language before it becomes a public knowledge claim.

Overview

Students write different captions for the same graph and analyze how language changes what the data seems to show.

Student Prompt

Inspect Graph Caption Framing Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Graph Caption Framing Stimulus A	Student-facing stimulus 1: Line graph: screen time rises during exam week and falls afterward. Annotate the wording. Circle the words that change tone, emotion, implication, or trust.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Graph Caption Framing Stimulus B	Student-facing stimulus 2: Caption contrast: 'Students lose balance during exams' versus 'Temporary study routines shift during assessment week'. Annotate the wording. Circle the words that change tone, emotion, implication, or trust.	Use this for comparison, a second round, or a partner challenge.
Graph Caption Framing Reveal / Twist	Student-facing stimulus 3: Caption contrast: 'Students lose balance during exams' versus 'Temporary study routines shift during assessment week'. Annotate the wording. Circle the words that change tone, emotion, implication, or trust.	Teacher reveal: connect the changed response to the big idea — Data needs language before it becomes a public knowledge claim.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Graph Caption Framing Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Graph Caption Framing Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	When does language become part of evidence?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how data needs language before it becomes a public knowledge claim.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Use a graph simple enough that students can understand it quickly.
- Remove the original title and caption so students experience the framing gap.

Ready-to-Use Examples

- Line graph: screen time rises during exam week and falls afterward.
- Caption contrast: 'Students lose balance during exams' versus 'Temporary study routines shift during assessment week'.

Suggested Timing

Stage	Teacher move	Watch for
1	Show the graph without title or caption and ask students what it might show.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Groups write captions in assigned framing styles.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Students compare how each caption changes interpretation and trust.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Groups identify what extra context the graph needs.	Assumptions, confidence shifts, evidence claims, and language choices.
5	Debrief the difference between data, interpretation, and persuasion.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- The graph is not silent; its labels teach us how to read it.
- What does your caption invite the reader to conclude?

Common Misconceptions

- Students may say language merely expresses thought rather than shaping interpretation.
- Students may identify bias in extreme examples but miss subtle framing.

Assessment Look-Fors

- Explains how wording changes interpretation before evidence is judged.
- Uses concrete language choices as evidence for a TOK claim.

Differentiation and Adaptations

- Support: provide caption sentence starters.
- Challenge: include axis-scale manipulation and ask students to detect it.

Extension Tasks

- Students rewrite a real graph caption to make it more cautious and transparent.
- Connect graph captions to headlines and public policy claims.

Related Activities

- Loaded Headline Laboratory
- Survey Wording Experiment
- Proof versus Pattern